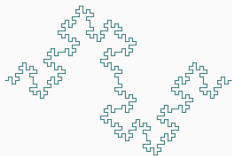


Course on Fair Allocation of Indivisible Goods

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Why we sometimes need less centralized mechanisms

There are many reasons why simpler mechanisms may still be worth studying:

- in restricted settings they may offer similar (or even stronger) guarantees,
- lack of access to (or trust in) a central authority – humans prefer to take part in the allocation process, to use simple, easy to understand, protocols
- may offer interesting compromise between communication burden and efficiency/fairness guarantees

Two agents with additive utilities: The Adjusted Winner

The protocol is designed for two agents, who initially have the same amount of points to assign to items.

It runs in two phases:

1. **winning phase**: allocate goods efficiently, ie. assign each good to the agent who values it most
2. **adjusting phase**: goods are transferred from the “high” agent to the “low” agent in increasing order of the ratio

$$\frac{u_h(r)}{u_l(r)}$$

until the poorest become the richest (or they enjoy the same utility).

Two agents with additive utilities: The Adjusted Winner

Two agents have 20 points to allocate.

						
agent 1	1	2	5	3	8	1
agent 2	2	3	9	1	2	3

Two agents with additive utilities: The Adjusted Winner

						
agent 1	1	2	5	3	8	1
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Winning phase

agent 1 enjoys utility 11

agent 2 enjoys utility 17

Two agents with additive utilities: The Adjusted Winner

						
agent 1	1	2	5	3	8	1
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Adjusting phase

$((\text{soccer ball}, \frac{3}{2}), (\text{umbrella}, \frac{9}{5}), (\text{diamond}, 2), (\text{bicycle}, 3))$

 must be transferred

Two agents with additive utilities: The Adjusted Winner

						
agent 1	1	2	5	3	8	1
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Splitting phase

If agents could split some part of ,

both agents could get exactly the same utility in the end.

Two agents with additive utilities: The Adjusted Winner

The protocol may thus require the last resource r to be split.

The idea is to split precisely so as to attain exactly the same utility for both agents. The “low” agent must get the following fraction:

$$\frac{u_h(r) + u_h(\pi \setminus \{r\}) - u_l(\pi \setminus \{r\})}{u_h(r) + u_l(r)}$$

However, without knowing in advance which resource may be splitted, it must be assumed that all are. Under this assumption:

Adjusted Winner returns an EF + PO allocation, and both agents enjoy the same utility.

Discrete versions (eg. by giving the last object to the agent who should get the largest part) guarantee EF1+PO.

Two agents with ordinal preferences: Simultaneous protocols

Assume ordinal preferences. (How shall we define EF?)

The "sequential" protocol of Brams and Taylor (BT) uses a **contested pile (CS)** of objects not allocated. It works as follows: while not (all the objects are assigned or put in the CS)

- agents A and B name their preferred objects among the remaining ones. If they differ, they get assigned those objects, otherwise they go in the CP

Note that it may generate incomplete allocations.

agent 1:						
agent 2:						

Many variants of this basic protocol exist.

Brams, Kilgour, Klamler. *Two-Person Fair Division of Indivisible Items: An Efficient, Envy-Free Algorithm*. Notices of the AMS.

Any number of agents with additive utilities: Picking sequences

We fix beforehand a sequence of agents, eg. $(n = 3, m = 6)$

[123231]

- agents pick one resource at a time, at their turn
- if they do so sincerely they pick the best resource available to them at that stage of the protocol

Only requires to **communicate** m times which resource to pick ($\log(m)$ bits), hence overall $m \log m$ bits.

Any number of agents with additive utilities: Picking sequences

Sequence = [123231]

						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

Any number of agents with additive utilities:

Picking sequences

Assuming for the moment that $k = m \bmod (n)$, ie. we can ensure that each agent gets the same number of resources.

Take a permutation of agents:

$$p = [p(1), p(2), \dots, p(n)]$$

Let p^{-1} be the “mirror” sequence of p .

- **round robin**: the subsequence p is repeated k times
- **balanced**: $(p \circ p^{-1})$ is repeated $k/2$ times

(When there are only two agents, it is common to talk about **strict alternation** or **balanced alternation**)

Brams and Taylor. *The Win-win Solution. Guaranteeing Fair Shares to Everybody*. 2000.

Bouveret and Lang. *A general elicitation-free protocol for allocating indivisible goods*. IJCAI-11.

Any number of agents with additive utilities: Picking sequences (Round-Robin)

Round-robin sequences are arguably the simplest ones (they are also called **draft mechanisms**).

When $n = m$ similar to **serial dictatorship**.

Do they have interesting properties?

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Round-robin picking sequences guarantee EF1

👉 during each k phase, when picking its resource r , agent i prefers r over the $n - 1$ ones subsequently chosen by other agents. Envy towards j can result from resource chosen by j before his first pick (during the first phase). Removing this resource from bundle of j removes envy.

Any number of agents with additive utilities: Picking sequences (Round-Robin)

What about Pareto-efficiency?

Any number of agents with additive utilities: Picking sequences (Round-Robin)

What about Pareto-efficiency?

						
agent 1	18	8	1	1	1	1
agent 2	4	6	5	5	5	5

Round-robin gives  to agent 1,  to agent 2, and two books each, yielding utilities = (20, 16).

Any number of agents with additive utilities: Picking sequences (Round-Robin)

What about Pareto-efficiency?

						
agent 1	18	8	1	1	1	1
agent 2	4	6	5	5	5	5

But exchanging  that agent 2 got against the two books from agent 1 obtained gives (26, 20).

Any number of agents with additive utilities: Picking sequences (Round-Robin)

What about Pareto-efficiency?

						
agent 1	18	8	1	1	1	1
agent 2	4	6	5	5	5	5

Round-robin picking sequences do not guarantee PO

Any number of agents with additive utilities: Picking sequences: Which are the fairest ones?

Under assumptions on preference distributions, we may look for sequences which maximize the **expected** efficiency/fairness measure. Suppose we randomly draw Borda utilities (orders are all equally probable, and converted into Borda utilities)

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2 agents: which sequence gives the better ESW on average?

✎ A: [122121] or B: [121221]

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3 agents and 5 goods: which sequence has highest ratio of PROP?

✎ Propose your own sequence or vote for one!

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3 agents and 5 goods: which sequence has highest ratio of PROP?

✎ Propose your own sequence or vote for one!

3 agents and 8 goods: which sequence has highest ration of EF?

✎ Propose your own sequence or vote for one!

Any number of agents with binary submodular utilities: Yankee Swap

Call an item useful if it brings positive marginal utility.

The protocol works in round, items initially unassigned. At each turn, consider the agent with lowest utility from A . Check whether the agent can either :

- (i) pick a useful unallocated item, or
- (ii) steal a useful item from an agent, if this agent can in turn pick or steal a useful item, etc.

If not, the agent is removed from A . Stop when A is empty.

Under binary submodular utilities, Yankee Swap guarantees a non-redundant (any item dropped decreases utility) Lorenz allocation, also Max-NW, Max-USW, EFX, LEX, $1/2$ -MMS.

One object per agent: House allocation

Under the hypotheses:

- ordinal preferences
- resources are not initially owned by agents

⇒ **Serial dictatorship**

- agents are ranked in a predefined order
- at each turn, each agent picks her favorite resource among the remaining ones
- what are the properties of this algorithm?

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- what are the properties of this algorithm?

⇒ Pareto-optimality and strategy proofness

Interestingly, **any** PO allocation can be obtained through an order of Serial Dictatorship

One object per agent: House allocation (example)

agent 1:					
agent 2:					
agent 3:					
agent 4:					
agent 5:					

House market: Top Trading Cycle

Under the hypotheses:

- ordinal preferences
- each agent initially owns a resource

The **Top Trading Cycle (TTC)** proceeds as follows:

- build a bipartite graph agents/resources
- each agent “points” towards her preferred resource
- each resource points to the agent owning it
- select one cycle, remove the agents and resources involved from the process, and repeat

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What are the properties of this protocol?

TTC is PO, Strategy-proof, et Individually Rational

House market: Top Trading Cycle

The allocation returned by TTC is the unique allocation of the **core** A^*

There is no coalition of agents X and allocation A' such that:

1. $\cup_{x \in X} A'(x) = \cup_{x \in X} A_0(x)$,
2. $A'(x) \succeq A^*(x)$ for all $x \in X$,
3. $A'(x) \succ A^*(x)$ for at least one $x \in X$.

Any number of agents and monotone utilities: Envy cycles

We first present informally the approach, based on a simple sequential allocation of resources.

For each resource r_k to be allocated:

- build the **envy graph** $G = (\mathcal{N}, E)$, where $(i, j) \in E \times E$ if agent i envies agent j
- while the graph has **cycles**, pick one $C = (c_1, c_2, \dots, c_q)$, and reallocates the bundle of c_i to c_{i-1} (and of c_1 to c_q).
- allocate r_k to an agent that **no one envies**.

Lipton et al. *On approximately fair allocations of divisible goods*. EC-2004.

Any number of agents and monotone utilities: Envy cycles

1

2

3

						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

Any number of agents and monotone utilities: Envy cycles

1

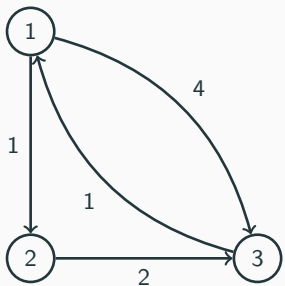
2

3

						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

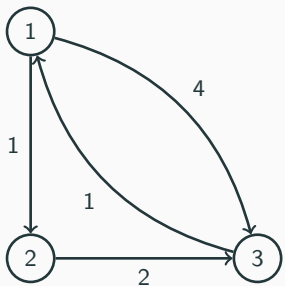
No object is allocated yet.

Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

There are two cycles: (1,3) or (1,2,3)

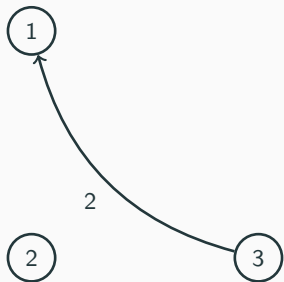
Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

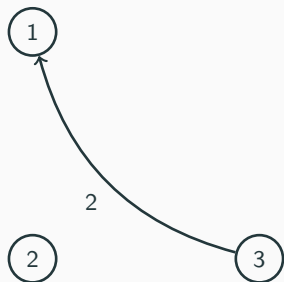
Suppose we chose cycle (1,2,3). After a single rotation, agent 1 and agent 2 are not envied any longer.

Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

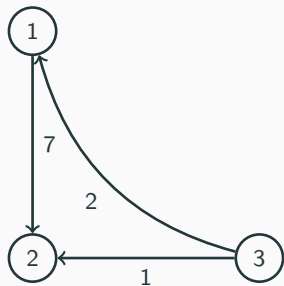
Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

We can give r_3 to agent 1. There are no cycle, agent 2 and agent 3 are not envied.

Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

We can give r_4 to agent 2. There are no cycles but only agent 3 is not envied.

Any number of agents and monotone utilities: Envy cycles



						
agent 1	1	2	5	3	7	2
agent 2	2	6	8	1	1	2
agent 3	5	4	4	3	2	2

We finally give r_5 to agent 3. The final allocation is not envy-free, as agent 1 envies agent 2.

Envy cycles: analysis

Cycle reallocation step: $C = (c_1, c_2, \dots, c_q)$

☞ Envy must have decreased.

- any agent in the cycle has increased its utility.
- bundles are unaffected

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Cycle reallocation step: $C = (c_1, c_2, \dots, c_q)$

☞ Envy must have decreased.

- any agent in the cycle has increased its utility.
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☞ The number of edges in the envy graph has decreased.

- edges between agents $\notin C$ are not affected
- edges from agents $\notin C$ to C now point to previous agent in C
- edges from agents $\in C$ to agents $\notin C$ may only decrease
- (original) edges between agents $\in C$ are deleted

Envy cycles: analysis

The envy cycle algorithm returns an EF1 allocation

Envy cycles: analysis

The envy cycle algorithm returns an EF1 allocation

Let α be the max value that any agent gives to a good.

Base case:

A_0 : allocate first resource randomly. Clearly $e(A_0) \leq \alpha$.

Induction step:

Suppose A with $\{r_1, \dots, r_k\}$ allocated, and $e(A) \leq \alpha$.

By repeatedly applying cycle reallocation in the envy graph, we must get an acyclic graph.

Hence at least an agent j is not envied: she gets r_{k+1} .

Envy among agents $\neq j$ is not affected.

Envy of agents $i \neq j$ towards j is $\leq \alpha$, since j was not envied.

Envy cycles: analysis

- **computational complexity** : cycle detection $O(n^2)$ and edge removing. At most n edges added by one resource, so overall (for all resources) nm edges to delete, hence $O(mn^3)$.
- **communication requirement**: can be implemented with $O(mn)$ values queries (but observe that the protocol does actually not require agents to communicate utilities)

Oh1, Procaccia, Suksompong. *Fairly Allocating Many Goods with Few Queries*. AAAI-2019.

Local Exchanges

We conclude with a **fully distributed** approach:

- resources are **initially** held by agents
- agents agree on **local rational deals**
- agents may have restrictions on the **types of deals** they can perform
- agents may not be able too see/deal with any other agents

This approach relies on a dynamics, with agents encountering each others and (potentially) agreeing on deals. The final allocation is when no more deals are possible.

Sandholm. *Contract types for satisficing task allocation*. AAAI Spring Symposium.

Endriss et al. *Negotiating socially optimal allocations of resources*. JAIR-06.

Local Exchanges

The notions of fairness/efficiency behave differently wrt. this distributed setting:

Intuitively:

- if some agents perform a deal which increase locally the **sum** of utilities, then globally the sum of utility will increase

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Local Exchanges

The notions of fairness/efficiency behave differently wrt. this distributed setting:

Intuitively:

- if some agents perform a deal which increase locally the **sum** of utilities, then globally the sum of utility will increase
- if some agents perform a deal which increase locally the **min** of utility, then globally the min of utility cannot decrease
- if some agents perform a deal which decrease locally **envy**, then globally envy may very well increase

This has consequences on convergence guarantees that can be given.

Let us illustrate this approach on the simple scenario of house markets:

- Reminder: same number of resources as agents, each agent can hold only hold one resource
- TTC is the method of choice with nice properties
- but suppose agents can simply perform rational **swap deals**

Damamme et al. *The power of swap deal in distributed resource allocation*. AAMAS-15.

Example

	5	4	3	2	1
agent 1:					 
agent 2:				 	
agent 3:					 
agent 4:					 
agent 5:		 			

Example

	5	4	3	2	1
agent 1:					 
agent 2:			 		
agent 3:			 		
agent 4:				 	
agent 5:		 			

Example

	5	4	3	2	1
agent 1:					 
agent 2:			 		
agent 3:			 		
agent 4:	 				
agent 5:		 			

Example

	5	4	3	2	1
agent 1:					
					
agent 2:					
					
agent 3:					
					
agent 4:					
					
agent 5:					
					

Example

	5	4	3	2	1	
agent 1:						
	▲					
agent 2:						
		▲				
agent 3:						$\Rightarrow sw_u(A) = 22$
		▲				$\Rightarrow sw_e(A) = 4$
agent 4:						
	▲					
agent 5:						
		▲				

What are the properties of such a protocol?

- Is Pareto-optimality guaranteed?
- What is the “price” of using this protocol wrt. egalitarian social welfare?
- What is the “price” of using this protocol wrt. number of pairwise envies? (ie. utilitarian social welfare in this case...)
- What is the complexity of the **reachability** question?

Is Pareto-optimality guaranteed?

Is Pareto-optimality guaranteed? No.

Is Pareto-optimality guaranteed? No.

Domain restriction guaranteeing Pareto-optimal outcomes?

Is Pareto-optimality guaranteed? No.

Domain restriction guaranteeing Pareto-optimal outcomes?

👉 *In a single-peaked domain, any sequence of rational swap deals reaches a Pareto-optimal allocation.*

Price for utilitarian social welfare / number of pairwise envies:

Local Exchanges

Price for utilitarian social welfare / number of pairwise envies:

Price is at most 2. Take a swap-stable allocation A : for each pair of agents (x, y) , at least one agent ranks the resource of the other below her current. Hence overall at least $n(n - 1)/2$ resources ranked below.

Local Exchanges

Price for utilitarian social welfare / number of pairwise envies:

Price is at most 2. Take a swap-stable allocation A : for each pair of agents (x, y) , at least one agent ranks the resource of the other below her current. Hence overall at least $n(n - 1)/2$ resources ranked below.

Price can be 2:

a_1	:	1	$\succ$	2	$\succ$	3	$\succ$	4	$\succ$	5
a_2	:	2	$\succ$	3	$\succ$	4	$\succ$	5	$\succ$	1
a_3	:	3	$\succ$	4	$\succ$	5	$\succ$	1	$\succ$	2
a_4	:	4	$\succ$	5	$\succ$	1	$\succ$	2	$\succ$	3
a_5	:	5	$\succ$	1	$\succ$	2	$\succ$	3	$\succ$	4

More on distributed settings

Other typical results in such settings:

- allowing the use of **money** and characterizing convergence properties under various protocols/preference constraints
- accounting for the underlying **visibility/deal graph**
- **communication complexity** (typically in terms of number of deals) of such protocols

Chevaileyre et al. *Allocating Goods on a Graph to Eliminate Envy*. AAAI-07.

Dunne. *Extremal behaviour in multiagent contract negotiation*. JAIR-05.

Other recent (and less recent) topics

Other topics

- manipulability of fair division mechanisms
- fair division with ordinal preferences
- online fair division
- groupwise notion of fairness
- fair division of goods and chores
- fair division on networks
- ...

Thank you!